PHYS 5041: Course Content

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Differential Equations I, `diffeq-I.pdf`

- Terminology: independent / dependent variables, linear / non-linear, order, homogeneous / inhomogeneous.
- Solving first-order ODEs: separable equations, linear equations (integrating factor), exact differentials, change of variables.
- Solving second-order ODEs: homogeneous equations with constant coefficients, Euler's Equation, inhomogeneous equations, homogeneous boundary conditions.
- Green Function.
- Hypergeometric Differential Equation, ${}_2F_1(a, b, c; x)$.
- Dirac Delta Function.
- Differentiating definite integrals.

Series, `series.pdf`

- Geometric series, binomial series.
- Convergence of series: preliminary test, comparison test, ratio test, integral test, alternating series, absolute / conditional convergence.
- Power series, Taylor series, L'Hôpital's rule.

Differential Equations II, `diffeq-II.pdf`

- Wronskian and its use for finding the second solution.
- Solution of second-order ODEs via generalized power series (Method of Frobenius), singular points, Fuch's Theorem, indicial equation, Frobenius method solution of the Hypergeometric Differential Equation.
- Green Function.

Orthogonal Functions, `orthogonal-functions.pdf`

- Orthogonality, completeness.
- Sturm-Liouville ODE, boundary conditions.
- Inner product.
- Expanding the Green Function in orthogonal functions.

Partial Differential Equations, `pdes.pdf`

- Identify named equations.
- Separation of variables technique.
- Matching boundary conditions with the help of orthogonal functions.

Fourier Series and Transforms, `fourier.pdf`

- Fourier series, exponential and trigonometric forms. Even and Odd functions.
- Parseval's theorem for Fourier series.
- Fourier transform, Parseval's theorem for Fourier transforms, convolution.
- Discrete Fourier transforms, convolution.